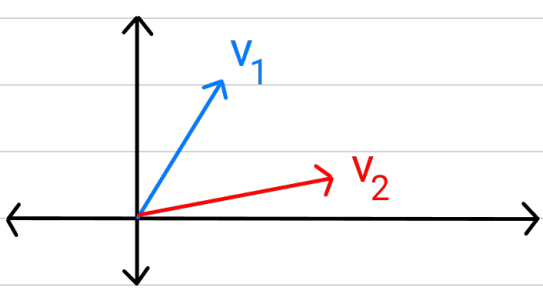
1.3 Vectors

Vectors are a part of vector spaces which become important later in this course (Chapter 4 and beyond). Also very important in higher level math. For now just the early ideas, especially those which relate to what we've studied/are studying with matrices.

Consider in $\mathbb{R}^2$ two vectors $v_1 = (1, 3)$, $v_2 = (2, 1)$

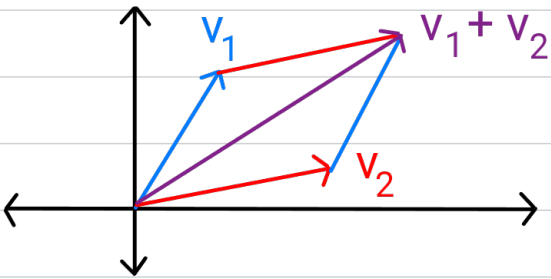
Geometrically can be expressed as follows:



That is, as rays starting at origin. Now consider

$$v_1 + v_2 = (1, 3) + (2, 1) = (3, 4)$$

Geometrically how does this look?



Completes parallelogram

So that's how adding two vectors works both mathematically (component wise) and geometrically

What about something a bit more extensive?

$$2v_1 + 3v_2 = 2(1, 3) + 3(2, 1) = (8, 9)$$

This is a <u>linear combination</u> of v_1 and v_2.

More generally, in any $\mathbb{R}^m$ for any set of vectors $\{v_1, v_2, \ldots, v_n\}$ a vector of the type:

$$c_1 v_1 + c_2 v_2 + \ldots + c_n v_n$$

is a linear combination of that set of vectors where the c_i's are all real constants.

We are often interested in knowing if we take the full set of linear combinations of a set of vectors, that is all possible choices for every c_i, what does that create? Does it generate the full space $\mathbb{R}^m$?
And if not then what?

How about this for the v_1, v_2 mentioned earlier?
Well two vectors which are not multiples of each other typically create a plane so we suspect those create the full space $\mathbb{R}^2$.

But if we added a third coordinate to each (arbitrary) would they then create $\mathbb{R}^3$? E.g. say

$$w_1 = (1, 3, 1) \quad , \quad w_2 = (2, 1, 5)$$

Well presumably those still create only a plane so we think no? And if they do only create a plane how can we describe it? Maybe by finding its equation?

To help with this we often look at vectors as matrices.

$$v_1 = (1, 3) = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 3 \end{bmatrix}$$

That is as a 2 x 1 or 1 x 2 matrix.

More generally a vector in $\mathbb{R}^m$ can be written as a m x 1 or 1 x m matrix (usually the former in this class for consistency).

The big question is: For m, n any positive integers, if we have a set of n vectors in $\mathbb{R}^m$ do they "create" the full space $\mathbb{R}^m$? And if not what do they create and how can you describe it?

By create we mean the set of all linear combinations of that set of vectors which is called the <u>span</u> of that that set of vectors.

Span is an important idea in general with vectors. One relevant reason is if you have a "piece" of $\mathbb{R}^m$, presumably with infinitely many vectors, how would you describe to someone, i.e. show what vectors are in that piece versus not?

Since infinitely many, you obviously can't list them one by one. If possible though, sometimes you can say it's the span of some small set of vectors, describes it efficiently even though it's an infinite set.

This is sort of similar to describing a function from $\mathbb{R}$ to $\mathbb{R}$. Again can't list points one by one but saying something like $f(x) = x^2$ imparts all that information efficiently.

If these n vectors are denoted $\{v_1, v_2, \ldots, v_n\}$, create an m x n matrix A whose columns are these vectors, that is

$$A = \begin{array}{c} \begin{bmatrix} \boxed{\begin{matrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{matrix}} & \boxed{\begin{matrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{matrix}} & \begin{matrix} \cdots \\ \cdots \\ \\ \cdots \end{matrix} & \boxed{\begin{matrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{matrix}} \end{bmatrix} \\ \begin{matrix} v_1 & \quad v_2 & \qquad\qquad v_n \end{matrix} \end{array}$$

This is sort of a standard way to write an m x n matrix, that is denoting the entries as a_{ij} where i represents the row number, and j is the column number.

But again the columns of A are the vectors in our set. Matrix A essentially is a "catalog" of our set of vectors.

Writing a set of vectors in this way is often useful both for cataloging purposes and because using things we know about matrices might help us better understand the set of vectors.

For instance, if we were to put A in RREF would that help us understand the set of vectors better? After all RREF is much easier to work with. Maybe so, but ok that would have to be carefully clarified, how might row operations affect things? Quite a lot to that.

1.4 Matrix Equations

The key thing is when we wish to know if some b in $\mathbb{R}^m$ is in the span of this set of vectors, that is equivalent to asking if the matrix equation

$$Ax = b$$

has a solution or not (i.e. is it a consistent linear system) Hence why this ties in with earlier ideas. Keep in mind these are matrices, not real numbers, don't say something like "let's divide by A", again it's a matrix, not a real number, you can't do that!

Let us clarify this notationally and also clarify why this equation being consistent or not is equivalent to whether or not b is in the span of our set of vectors.

This is matrix multiplication. Ok simplified version since x is allowed only one column but still a good way to

start getting acclimated to that as we will be doing that alot more in the next couple chapters.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n \end{bmatrix}$$

Each row of A "times" x, that's what we call <u>dot product</u> another idea that we will encounter quite a bit in this class, especially later on in vector spaces.

A different way to write this though is

$$x_1 v_1 + x_2 v_2 + \ldots + x_n v_n$$

That is, a linear combination of our set of vectors with the entries of x being the constant multiples (scalars).

The fact that Ax has this form shows why b being in the span of this set of vectors is equivalent to $Ax = b$ being a consistent system.

Thus to see if our set of vectors spans $\mathbb{R}^m$ we need to check for every b in $\mathbb{R}^m$ that $Ax = b$ is consistent.

So is this the same thing as what you did in 1.1/1.2?

Say we want to check if the set of vectors

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}, \begin{bmatrix} 9 \\ 10 \\ 20 \end{bmatrix} \right\}$$

spans $\mathbb{R}^3$ or not.

To that end, maybe we should check if this system

$$\begin{bmatrix} 1 & 5 & 9 \\ 2 & 6 & 10 \\ 3 & 7 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 1000 \end{bmatrix}$$

is consistent? That is, see if $\begin{bmatrix} 5 \\ 7 \\ 1000 \end{bmatrix}$ is in the span of those three? Would that give us the information we need?

Unfortunately not really. Problem is that only checks one single vector! If it happens to be in the span that doesn't prove every vector in $\mathbb{R}^3$ is! That's a a very important distinction to realize.

So this isn't exactly the same thing that we were doing in 1.1/1.2

In one sense it is similar since we are still checking for consistency, but there is also a big difference, that being that we want/need to check every b in $\mathbb{R}^m$, so not just one linear system, infinitely many systems at once.

So since we obviously can't check one by one, this is a very different/much harder question.

That's why for now we focus on the question of whether the span is the full $\mathbb{R}^m$ or not. The harder question of what the span is exactly when not the full space we will address more later when we have more tools.

To the question of whether the span is the full space the following theorem helps a lot. But first some useful notation.

<u>Notation:</u> $\forall$ means "for all"

$\in$ means "belongs" or "in"

Both of these are used alot in higher level math. And we will use them often in this class so you should be familiar with them.

<u>Theorem:</u> The following four statements are equivalent

(1) $\forall\ b \in \mathbb{R}^m$ the equation $Ax = b$ has a solution
(2) Each $b \in \mathbb{R}^m$ is a linear combination of the columns of A.
(3) The columns of A span $\mathbb{R}^m$
(4) A has a pivot position in every row.

Important again, if you were going to try to prove (1) was true, you can't just show it works for one b then claim it works for all! That's a common mistake people make.

More generally with true/false questions when showing something is false, can just give one counterexample. But when showing something is true you can't just show it works in one case then claim it works in all!

The first three of these statements stand out as being similar in part based on what we showed earlier with matrix multiplication.

The fourth statement is in a sense the key one, both because it is the one that doesn't seem as clear to be similar to the others and because if we want to check if any of these are true, that's the one we are best equipped to check based on prior ideas.

So why is (4) equivalent to the others? Say it is true.

Then since A has a pivot in every row, $m \leq n$ is assured as say a 5×3 matrix can't have five pivots since in any matrix each row has at most one pivot as does each column.

So then in the augmented matrix $[A \mid b]$, that is size $m \times (n + 1)$ so since still m rows, at most m pivots.

Therefore $[A \mid b]$ has the same m pivots as A so the additional column, the b vector, won't have a pivot, meaning $Ax = b$ is consistent always no matter what b is.

That proves one direction. For the other, suppose A is lacking a pivot in at least one row. Then in the RREF of A that row is a zero row. So choose b so that when we reduce $[A \mid b]$ that row will have a nonzero value in the appended column (since b can have any value there to start this is doable).

That will ensure $Ax = b$ is inconsistent since the augmented matrix has a pivot in the last column which proves the other direction.

Thus there is at least one b for which $Ax = b$ is inconsistent proving the equivalence to the other three statements.

1.5 Solutions to Linear Systems

So if we have some type of idea of which b the system $Ax = b$ is consistent for, we might want to know which x's are the solutions (inherently also is there a unique such x or many such x).

To help assess this, let's check the easiest case first:

$$Ax = 0$$

Remember these are matrices, 0 means 0 vector/matrix not the real number 0.

This is always consistent since $x = 0$ is always a solution (or can say the augmented matrix certainly can't have a pivot in the last column since it's all 0s).

So more interesting question is whether this has a <u>nontrivial</u> solution, i.e. a nonzero solution.

$Ax = 0$ is called a <u>homogeneous</u> equation. That term we encountered in Math 52 when solving second order differential equations with constant coefficients, e.g. for y a function of x, consider the two differential equations:

$$y'' + y = 0$$
$$y'' + y = e^x$$

The first equation is homogeneous since right side is 0 while second is nonhomogeneous since right side is not identically 0.

Big thing was solving nonhomogeneous relied on first solving homogeneous counterpart and then finding a single solution to nonhomogeneous equation perhaps via undetermined coefficients (will do this at end of this course too!).

As it turns out, the same type of thing will apply here. Solving $Ax = b$ for an arbitrary b will be aided by first fully solving $Ax = 0$. Hence one reason why we use the same terminology even though this is a very different context, i.e doesn't involve derivatives/calculus.